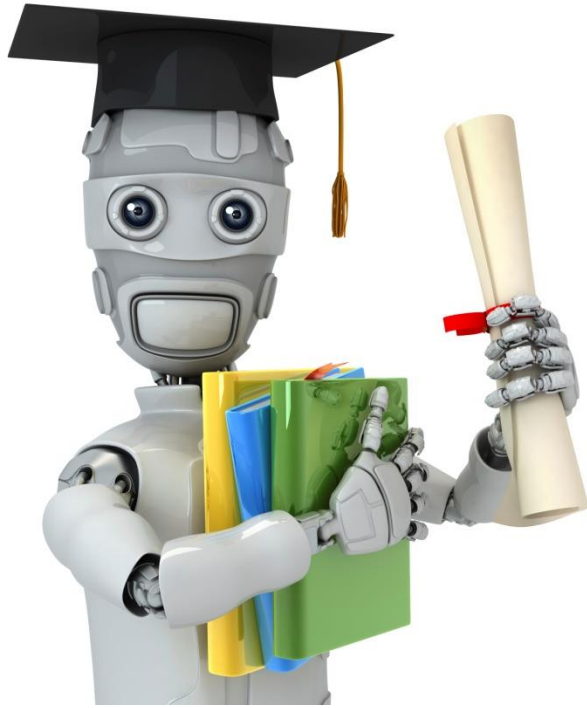




Machine Learning

# Linear Regression with multiple variables

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## Multiple features

## Multiple features (variables).

| Size (feet <sup>2</sup> ) | Price (\$1000) |
|---------------------------|----------------|
| $\rightarrow x$           | $y \leftarrow$ |
| 2104                      | 460            |
| 1416                      | 232            |
| 1534                      | 315            |
| 852                       | 178            |
| ...                       | ...            |

$$\underline{h_{\theta}(x) = \theta_0 + \theta_1 x}$$

# Multiple features (variables).

| Size (feet <sup>2</sup> ) | Number of bedrooms | Number of floors | Age of home (years) | Price (\$1000) |
|---------------------------|--------------------|------------------|---------------------|----------------|
| $x_1$                     | $x_2$              | $x_3$            | $x_4$               | $y$            |
| 2104                      | 5                  | 1                | 45                  | 460            |
| 1416                      | 3                  | 2                | 40                  | 232            |
| 1534                      | 3                  | 2                | 30                  | 315            |
| 852                       | 2                  | 1                | 36                  | 178            |
| ...                       | ...                | ...              | ...                 | ...            |

$m = 47$

Notation:

- $\rightarrow n$  = number of features  $n = 4$
- $\rightarrow x^{(i)}$  = input (features) of  $i^{th}$  training example.
- $\rightarrow x_j^{(i)}$  = value of feature  $j$  in  $i^{th}$  training example.

$$\underline{x^{(2)}} = \begin{bmatrix} 1416 \\ 3 \\ 2 \\ 40 \end{bmatrix}$$

$x_3^{(2)} = 2$

## Hypothesis:

Previously:  $h_{\theta}(x) = \theta_0 + \theta_1 x$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

e.g.  $\underline{h_0(x)} = \underline{80} + \underline{0.1x_1} + \underline{0.01x_2} + 3x_3 - 2x_4$   
↑ ↑ ↑  
age

$$\rightarrow h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

For convenience of notation, define  $x_0 = 1$ . ( $x_0^{(i)} = 1$ )

$$x = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$h_{\theta}(x) = \theta_0 x_0 + \theta_1 x_1 + \dots + \theta_n x_n$$

$$= \theta^T x$$

$$[\theta_0 \ \theta_1 \ \dots \ \theta_n]$$

$\theta^T$

(n+1) x 1  
matrix

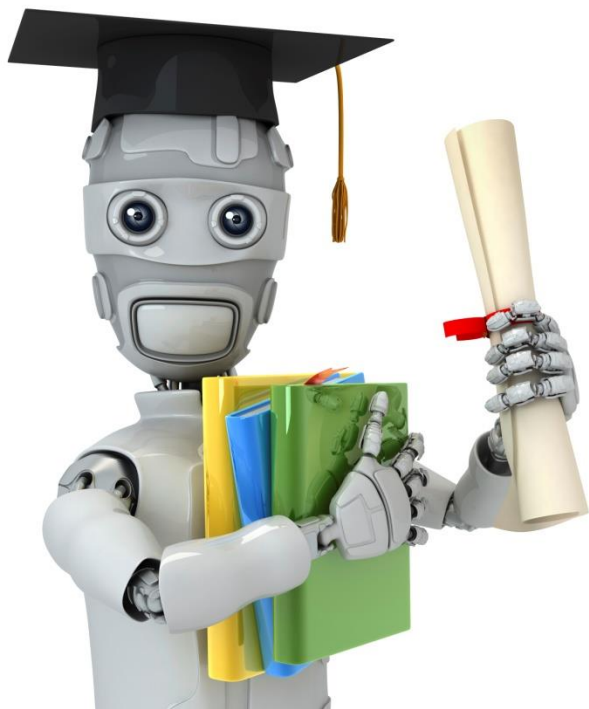
$$\theta^T x$$

$$\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$x$

Multivariate linear regression.  $\leftarrow$





Machine Learning

# Linear Regression with multiple variables

---

## Gradient descent for multiple variables

Hypothesis:  $\overset{\nearrow x_0=1}{h_{\theta}(x) = \theta^T x = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n}$

Parameters:  $\theta_0, \theta_1, \dots, \theta_n$   $\theta$   $n+1$ -dimensional vector

Cost function:

$$\underbrace{J(\theta_0, \theta_1, \dots, \theta_n)}_{J(\theta)} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

Gradient descent:

Repeat {

$\rightarrow \theta_j := \theta_j - \alpha \left[ \frac{\partial}{\partial \theta_j} \underbrace{J(\theta_0, \dots, \theta_n)}_{J(\theta)} \right]$   
↑  
(simultaneously update for every  $j = 0, \dots, n$ )



# Gradient Descent

Previously ( $n=1$ ):

Repeat {

→  $\theta_0 := \theta_0 - \alpha \underbrace{\frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})}_{\frac{\partial}{\partial \theta_0} J(\theta)}$

→  $\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \underbrace{x_1^{(i)}}_{\text{simultaneously update } \theta_0, \theta_1}$

}

New algorithm ( $n \geq 1$ ):

Repeat {

→  $\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$   
(simultaneously update  $\theta_j$  for  $j = 0, \dots, n$ )

}

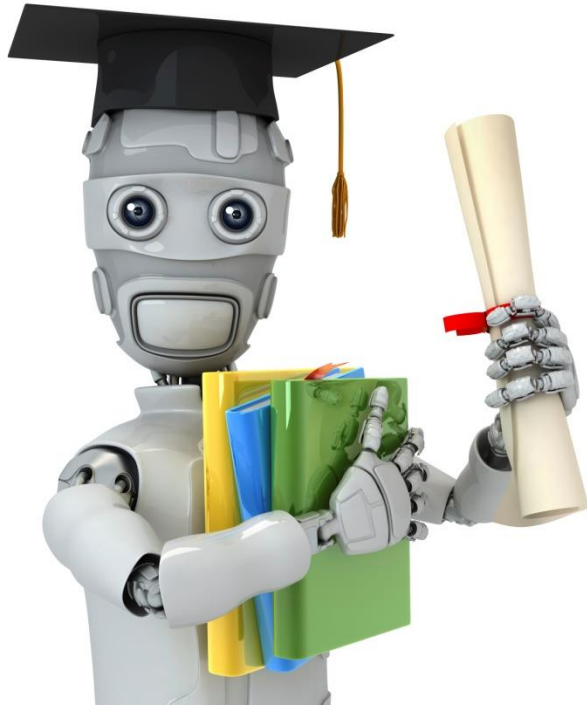
→  $\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \underbrace{x_0^{(i)}}_{x_0^{(i)} = 1}$

→  $\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_1^{(i)}$

→  $\theta_2 := \theta_2 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_2^{(i)}$

...





Machine Learning

# Linear Regression with multiple variables

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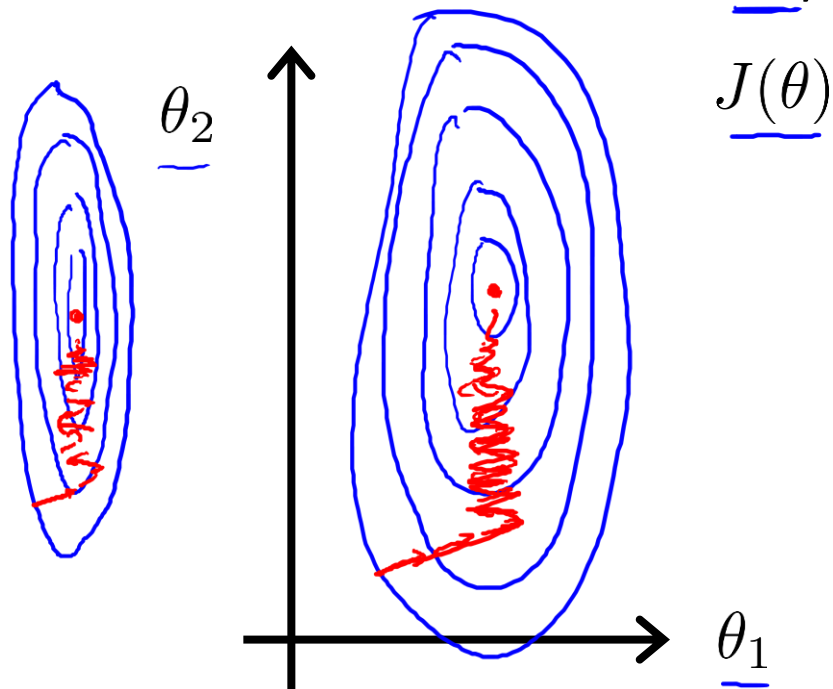
## Gradient descent in practice I: Feature Scaling

# Feature Scaling

Idea: Make sure features are on a similar scale.

E.g.  $x_1 = \text{size (0-2000 feet}^2\text{)}$  ←

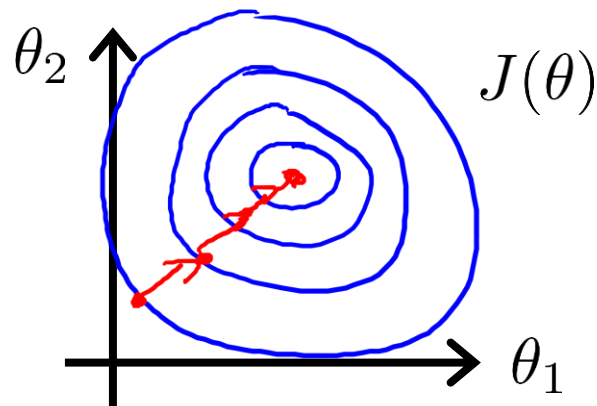
$x_2 = \text{number of bedrooms (1-5)}$  ←



$$\rightarrow x_1 = \frac{\text{size (feet}^2\text{)}}{2000} \quad \checkmark$$

$$\rightarrow x_2 = \frac{\text{number of bedrooms}}{5} \quad \checkmark$$

$$0 \leq x_1 \leq 1 \quad 0 \leq x_2 \leq 1$$



## Feature Scaling

Get every feature into approximately a  $-1 \leq x_i \leq 1$  range.

$$x_0 = 1$$

$$0 \leq x_1 \leq 3 \quad \checkmark$$

$$-2 \leq x_2 \leq 0.5 \quad \checkmark$$

$$-100 \leq x_3 \leq 100 \quad \times$$

$$-0.0001 \leq x_4 \leq 0.0001 \quad \times$$

$$-3 \text{ to } 3 \quad \checkmark$$

$$-\frac{1}{3} \text{ to } \frac{1}{3} \quad \checkmark$$

# Mean normalization

Replace  $x_i$  with  $x_i - \mu_i$  to make features have approximately zero mean  
(Do not apply to  $x_0 = 1$ ).

E.g.  $\rightarrow x_1 = \frac{\text{size} - 1000}{2000}$

Average size = 1000

$$x_2 = \frac{\# \text{bedrooms} - 2}{5}$$

1-5 bedrooms

$$\rightarrow [-0.5 \leq x_1 \leq 0.5], [-0.5 \leq x_2 \leq 0.5]$$

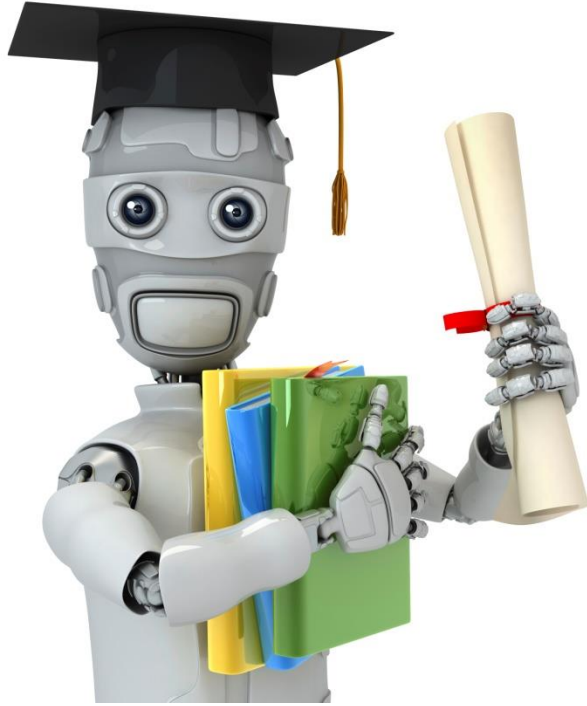
$$x_1 \leftarrow \frac{x_1 - \mu_1}{S_1}$$

← avg value of  $x_1$  in training set

range (max-min)  
(or standard deviation)

$$x_2 \leftarrow \frac{x_2 - \mu_2}{S_2}$$





Machine Learning

# Linear Regression with multiple variables

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Gradient descent in practice II: Learning rate

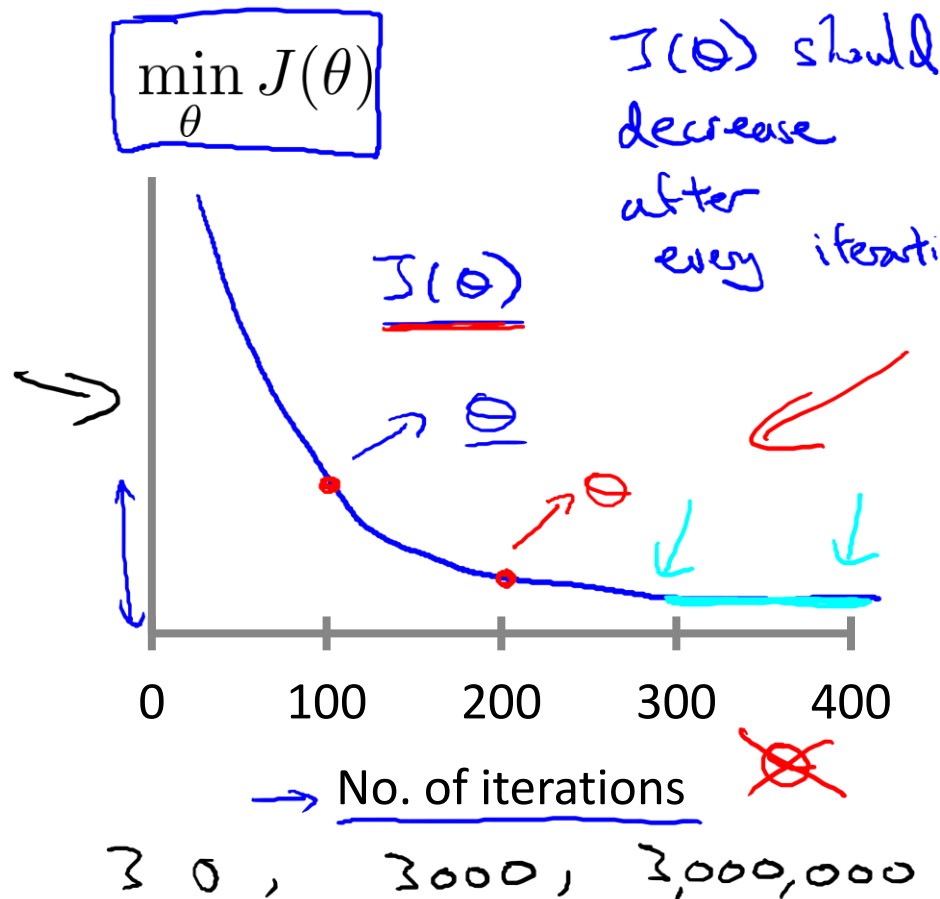


# Gradient descent

$$\rightarrow \theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

- “Debugging”: How to make sure gradient descent is working correctly.
- How to choose learning rate  $\alpha$ .

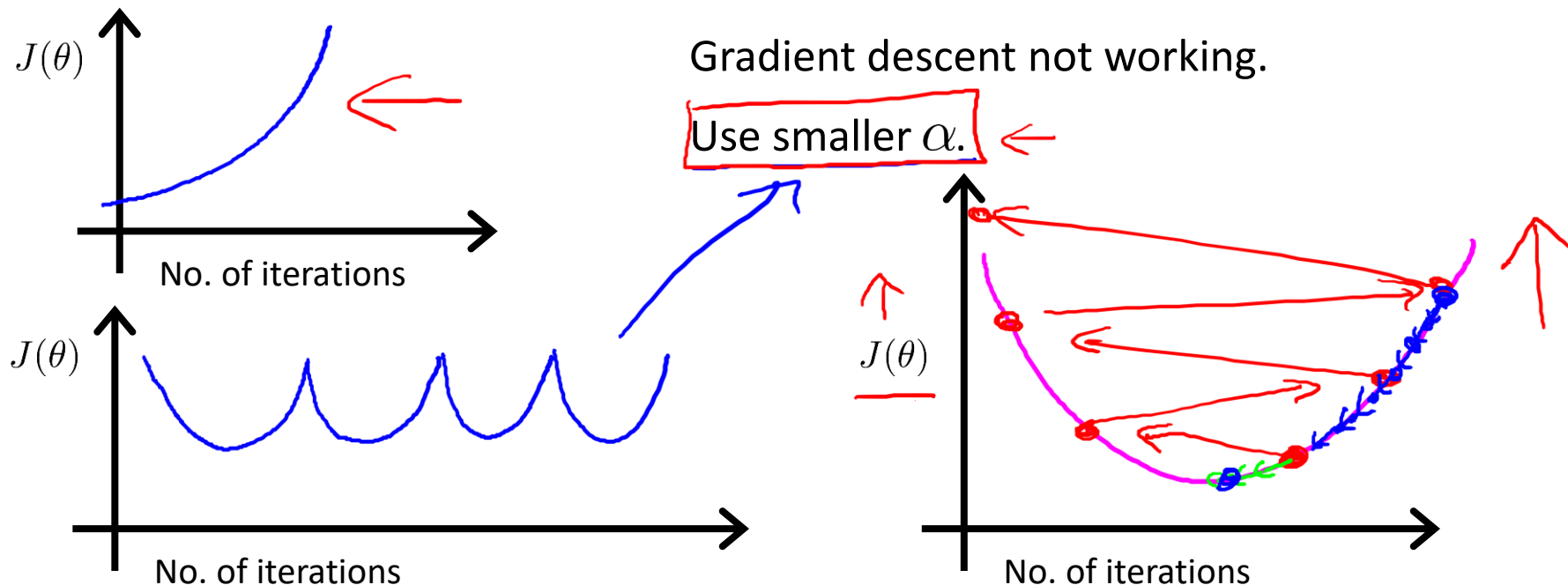
# Making sure gradient descent is working correctly.



→ Example automatic convergence test:

→ Declare convergence if  $J(\theta)$  decreases by less than  $10^{-3}$  in one iteration.

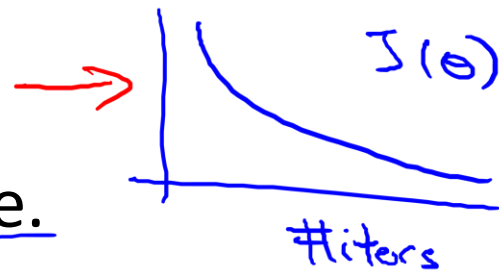
# Making sure gradient descent is working correctly.



- For sufficiently small  $\alpha$ ,  $J(\theta)$  should decrease on every iteration.
- But if  $\alpha$  is too small, gradient descent can be slow to converge.

## Summary:

- If  $\alpha$  is too small: slow convergence.
- If  $\alpha$  is too large:  $J(\theta)$  may not decrease on every iteration; may not converge.



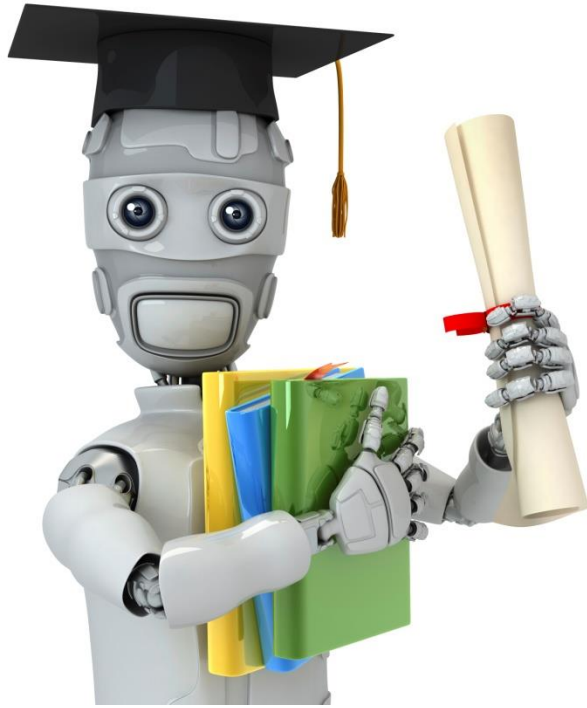
(Slow converge also possible.)

To choose  $\alpha$ , try

$\dots, \underline{0.001}, \underline{0.003}, \underline{0.01}, \underline{0.03}, \underline{0.1}, \underline{0.3}, \underline{1}, \dots$

Blue arrows and labels indicate a sequence of values:  $\uparrow$  (under 0.001),  $\nearrow 3\times$  (from 0.001 to 0.003),  $\nearrow \approx 3\times$  (from 0.003 to 0.01),  $\nearrow 3\times$  (from 0.01 to 0.03),  $\nearrow \approx 3\times$  (from 0.03 to 0.1),  $\nearrow$  (under 0.3), and  $\uparrow$  (under 1).





Machine Learning

# Linear Regression with multiple variables

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Features and  
polynomial regression

# Housing prices prediction

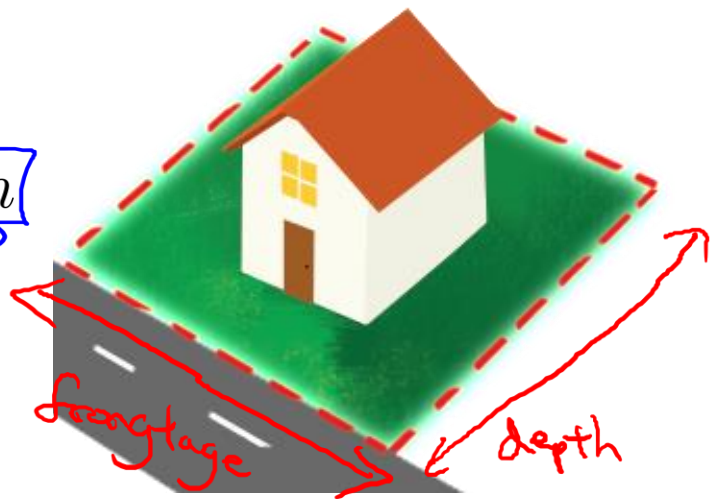
$$h_{\theta}(x) = \theta_0 + \theta_1 \times \underbrace{\text{frontage}}_{x_1} + \theta_2 \times \underbrace{\text{depth}}_{x_2}$$

Area

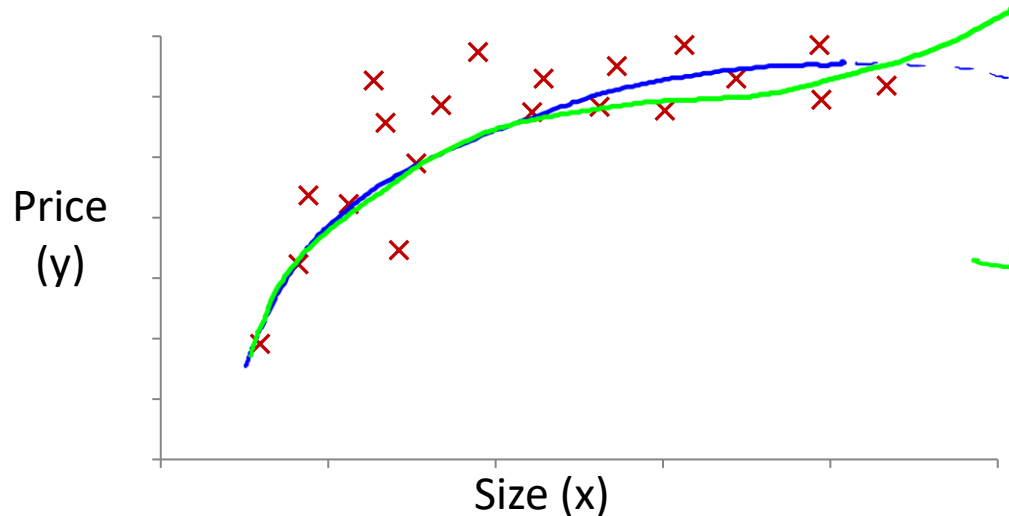
$$x = \underline{\text{frontage} * \text{depth}}$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

↖ land area



# Polynomial regression



$\theta_0 + \theta_1 x + \theta_2 x^2$

$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$= \theta_0 + \theta_1(\text{size}) + \theta_2(\text{size})^2 + \theta_3(\text{size})^3$$

$\rightarrow x_1 = (\text{size})$

$\rightarrow x_2 = (\text{size})^2$

$\rightarrow x_3 = (\text{size})^3$

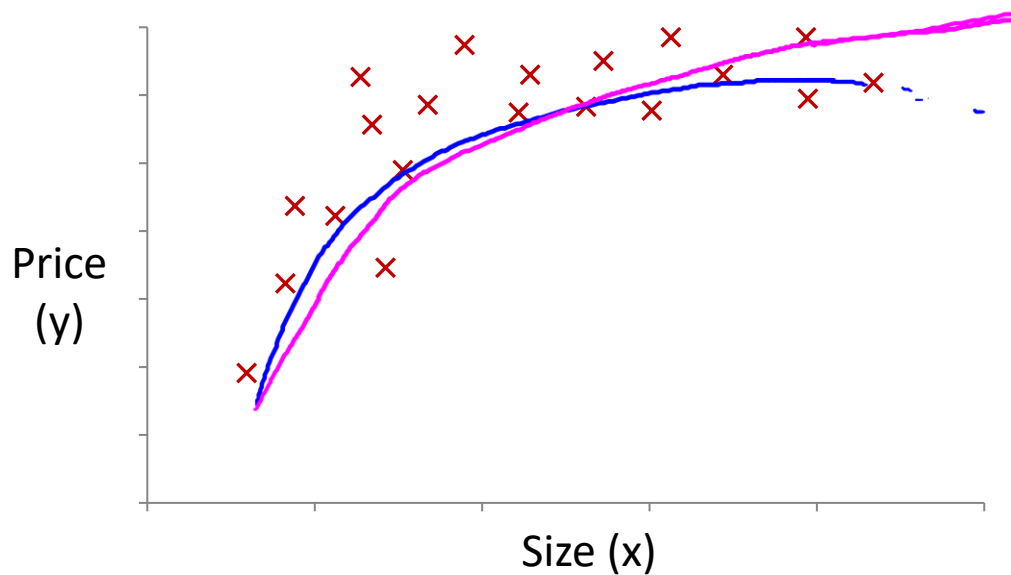
Size: 1-1000

Size<sup>2</sup>: 1-1,000,000

Size<sup>3</sup>: 1-10<sup>9</sup>



# Choice of features

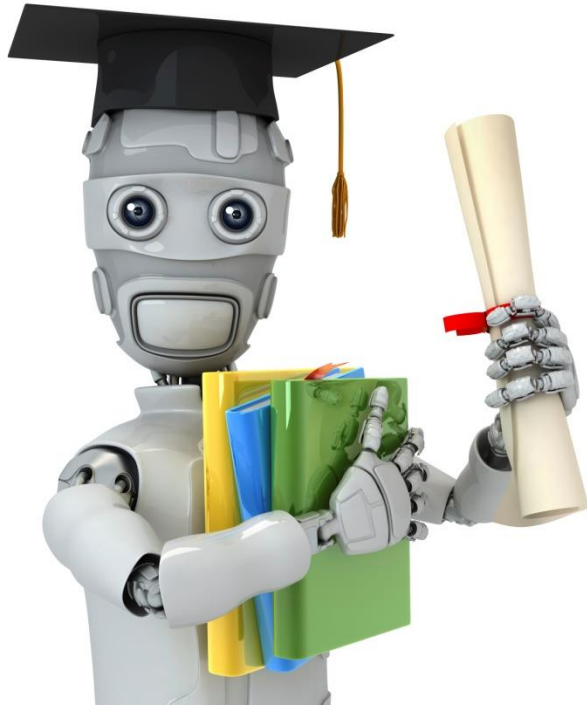


$$\rightarrow h_{\theta}(x) = \theta_0 + \theta_1(\text{size}) + \theta_2(\text{size})^2$$

$$\rightarrow h_{\theta}(x) = \theta_0 + \theta_1(\text{size}) + \theta_2\sqrt{(\text{size})}$$







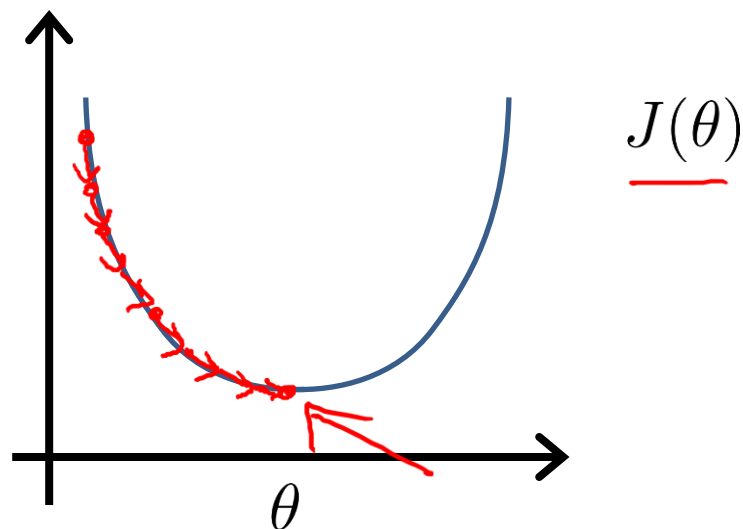
Machine Learning

# Linear Regression with multiple variables

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## Normal equation

# Gradient Descent



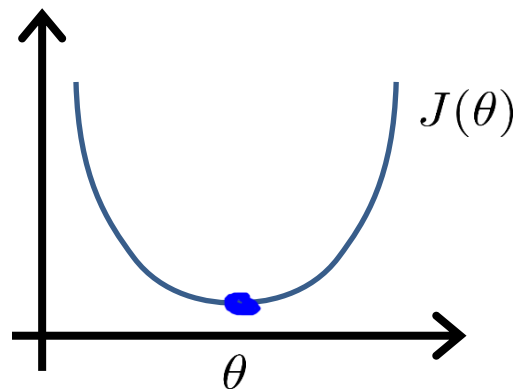
Normal equation: Method to solve for  $\theta$   
analytically.

Intuition: If 1D ( $\theta \in \mathbb{R}$ )

$\rightarrow$   $J(\theta) = a\theta^2 + b\theta + c$

$\frac{d}{d\theta} J(\theta) = \dots \stackrel{\text{set}}{=} 0$

Solve for  $\theta$



---

$\theta \in \mathbb{R}^{n+1}$        $J(\theta_0, \theta_1, \dots, \theta_m) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

$\frac{\partial}{\partial \theta_j} J(\theta) = \dots \stackrel{\text{set}}{=} 0$       (for every  $j$ )

Solve for  $\theta_0, \theta_1, \dots, \theta_n$

Examples:  $m = 4$ .

|       | Size (feet <sup>2</sup> ) | Number of bedrooms | Number of floors | Age of home (years) | Price (\$1000) |
|-------|---------------------------|--------------------|------------------|---------------------|----------------|
| $x_0$ | $x_1$                     | $x_2$              | $x_3$            | $x_4$               | $y$            |
| 1     | 2104                      | 5                  | 1                | 45                  | 460            |
| 1     | 1416                      | 3                  | 2                | 40                  | 232            |
| 1     | 1534                      | 3                  | 2                | 30                  | 315            |
| 1     | 852                       | 2                  | 1                | 36                  | 178            |

$X = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \end{bmatrix}$   
 $m \times (n+1)$

$y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$   
 $m$ -dimensional vector

$\theta = (X^T X)^{-1} X^T y$

$m$  examples  $(x^{(1)}, y^{(1)}), \dots, (x^{(m)}, y^{(m)})$  ;  $n$  features.

$$\underline{x^{(i)}} = \begin{bmatrix} x_0^{(i)} \\ x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_n^{(i)} \end{bmatrix} \in \mathbb{R}^{n+1}$$

$$X = \begin{bmatrix} \text{---} (x^{(1)})^T \text{---} \\ \text{---} (x^{(2)})^T \text{---} \\ \vdots \\ \text{---} (x^{(m)})^T \text{---} \end{bmatrix}$$

E.g. If  $\underline{x^{(i)}} = \begin{bmatrix} 1 \\ x_1^{(i)} \end{bmatrix}$   $\underline{X} =$

$$\Theta = (X^T X)^{-1} X^T y$$

$$\begin{bmatrix} | & X^{(1)}_1 \\ | & X^{(1)}_2 \\ \vdots & \vdots \\ | & X^{(1)}_m \end{bmatrix} \quad | \quad y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T y$$



$(X^T X)^{-1}$  is inverse of matrix  $X^T X$ .

Set  $A = X^T X$

$$(X^T X)^{-1} = A^{-1}$$

Octave: `pinv(X' * X) * X' * y`

$$\text{pinv}(X^T * X) * X^T * y$$

$$\theta = (X^T X)^{-1} X^T y$$

$\min J(\theta)$

$X'$

$X^T$

~~Feature Scaling~~

$$0 \leq x_1 \leq 1$$

$$0 \leq x_2 \leq 1000$$

$$0 \leq x_3 \leq 10^{-5}$$





$m$  training examples,  $n$  features.

### Gradient Descent

- • Need to choose  $\alpha$ .
- • Needs many iterations.
- Works well even when  $n$  is large.



$$\underline{n = 10^6}$$



### Normal Equation

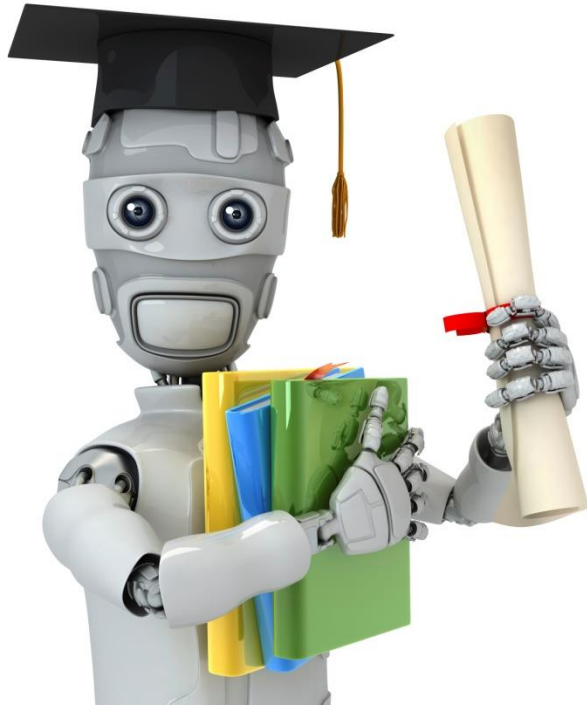
- • No need to choose  $\alpha$ .
- • Don't need to iterate.
- Need to compute
- •  $(X^T X)^{-1}$   $\frac{n \times n}{n \times n}$   $O(n^3)$
- Slow if  $n$  is very large.

$$n = 100$$

$$n = 1000$$

$$- - - \underline{n = 10000}$$





Machine Learning

# Linear Regression with multiple variables

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Normal equation  
and non-invertibility  
(optional)

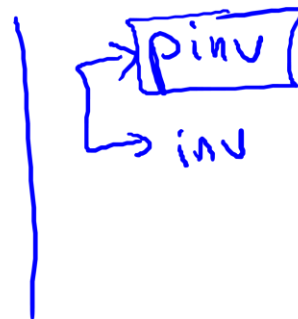
## Normal equation

$$\theta = \underline{(X^T X)^{-1} X^T y}$$

$$\underline{X^T X}$$

- What if  $\boxed{X^T X}$  is non-invertible? (singular/  
degenerate)

- Octave: `pinv(X' * X) * X' * y`



What if  $X^T X$  is non-invertible?

- Redundant features (linearly dependent).

E.g.  $x_1 = \text{size in feet}^2$

~~$x_2 = \text{size in m}^2$~~

$$x_1 = (3.28)^2 x_2$$

$$1\text{m} = 3.28\text{ feet}$$

$$\rightarrow m = 10 \leftarrow$$

$$\rightarrow n = 100 \leftarrow$$

$$\Theta \in \mathbb{R}^{101}$$

- Too many features (e.g.  $m \leq n$ ).

- Delete some features, or use regularization.

↓ later

